Empório da Música - customer service agent

Luccas Bassani | AI Engineer technical challenge - Artefact | 3 November 2026

Project repository

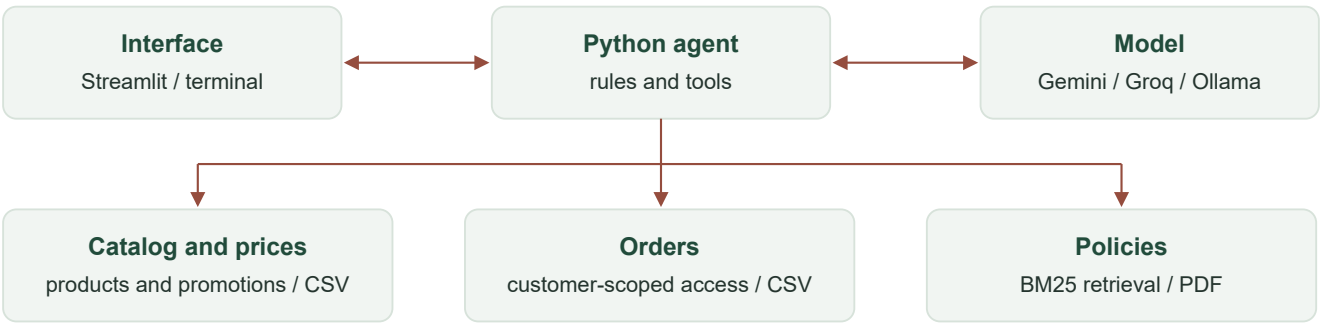
A Python prototype that answers questions about instruments, prices, availability, orders and store policies. I prioritized a local application that is straightforward to run, with traceable queries against the supplied data and real interaction examples.

Scope and persona

Lia serves customers in Brazilian Portuguese with a warm, concise and consultative tone. She should use the conversation context, ask one necessary question at a time and redirect unrelated requests. For returns and warranties, she guides customers towards staff review without approving requests herself.

I defined the interface reference, persona and scope. The interface supports agent configuration, chat, inspection of queries, and conversation/configuration exports. The Python implementation is independent of the reference used.

Architecture and technical decisions



- **Python and Streamlit:** interface and agent share one language, keeping setup simple for the demonstration.
- **Function calling and Python functions:** the model selects from five tools; Python validates arguments, queries CSVs and calculates prices with `Decimal`. Direct queries suit the supplied data volume without an additional database.
- **Lexical RAG with BM25:** the manual is small. Local retrieval returns up to four passages with section/page provenance, without requiring an embedding service.
- **Gemini Flash-Lite for the demonstration:** tool support and free-tier capacity observed during testing. Groq remains configurable; the Ollama connector is implemented but has not been validated with an installed local model.

The agent uses `httpx` without an agent framework to keep the tool request, execution and response cycle explicit. The prompt separates query rules, limits and sources from style preferences. Model context includes the previous two complete turns.

Setup and execution

Requires Python 3.12 or later. Verified environment: Windows with Python 3.14.4. In PowerShell, from the repository directory:

```
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.lock.txt
Copy-Item .env.example .env
```

Copy the example only during initial setup. In `.env`, select Gemini and enter your own key from Google AI Studio:

```
LLM_PROVIDER=gemini
GEMINI_API_KEY=your_key
GEMINI_MODEL=gemini-3.5-flash-lite

.\.venv\Scripts\python.exe -m streamlit run app.py
```

Open http://127.0.0.1:8501. Lia is already configured. For order tests, select **test customer #2** and ask about **order 8**; **customer #6** has **order 9**, without tracking information. Expand each answer to inspect queries and sources. The interface screenshot shows the flow.

Changing the customer, applying configuration or starting a new conversation clears the current history. Download buttons export conversations as Markdown/TXT and configuration as JSON. API keys remain in `.env`, outside Git and exports; restart the server after changing them. Existing process environment variables take precedence over the file.

Alternative providers

For Groq, set `LLM_PROVIDER=groq`, `GROQ_API_KEY` and `GROQ_MODEL=qwen/qwen3.8-27b`. For Ollama, install its server, run `ollama pull qwen3:8b`, and set `LLM_PROVIDER=ollama`, `OLLAMA_MODEL=qwen3:8b` and `OLLAMA_BASE_URL=http://127.0.0.1:11434`.

Provider selection is explicit, without automatic fallback. External APIs depend on account quotas. For Gemini's free tier, review the data-use terms. This demonstration uses the fictional challenge data and omits customer contact fields from order-tool results.

Tests and command line

```
.\.venv\Scripts\python.exe -m unittest discover -s tests -v
.\.venv\Scripts\python.exe -m emporio --help
```

Local tests need no API key or network. The CLI also supports catalog queries, policy retrieval and chat. `--check` and `--test-tools` make real diagnostic requests to the selected provider.

Data rules and validation

Availability requires active status and positive stock. An active promotion takes precedence over the PIX discount; discounts do not stack. Historical order totals are preserved. Data conflicts and policy gaps are recorded separately, without modifying the original sources.

The application fixes the test customer outside model arguments and restricts orders to that selection. Missing orders and another customer's orders receive the same guidance. Order tools return only necessary operational fields, excluding customer names and contact details.

61 automated tests passed, covering pricing, order access, retrieval, provider contracts and the interface. API responses are mocked in these tests. A real-API evaluation covered **32 scenarios: 12 passed, 15 passed with caveats and five failed**. After fixes, all five met their main objective in targeted retests, with caveats remaining in two. A shipping assertion regression in an order answer was also fixed and retested.

The full suite of 32 real scenarios was not rerun after the last fix. The evaluation report identifies each revision. There are five real examples, also included in Interaction examples and review.

Limitations and next steps

- Lexical retrieval can miss paraphrases; valid citations do not verify every claim. Tone, triage and price qualifications still vary across generations.
- Customer selection is a demonstration, without real authentication. CSVs are snapshots; the agent does not execute purchases, returns or live carrier queries.
- History is session-based, with two previous turns in model context. Ollama needs validation with a local model; external APIs depend on capacity and availability.

With more time, I would prioritize guardrails, structured triage and previous-interaction memory. Larger datasets would motivate hybrid retrieval and database/API queries. Fine-tuning would be a later experiment for tone and behavior using reviewed conversations and separate evaluation. See Improvements with more time.

AI assistance and project ownership

I used Codex to support implementation, test creation/execution and documentation. I directed the interface reference, persona, scope and provider choices, focusing on cost and simplicity.

I also guided test scenarios and expected outcomes, reviewed answers and prioritized fixes. When proposed approaches were too complex for the challenge, I directed their simplification. Development proceeded incrementally, using queries, answers and tests to review each stage.

Interaction examples and review

These five conversations were run on 3 November 2026 with `gemini-3.5-flash-lite`, Streamlit AppTest and the real API. Customer-facing dialogue is preserved in its original Portuguese as evidence of the Brazilian store experience. The surrounding submission documentation is in English. Linked exports retain the complete queries and sources.

1. Catalog and PIX follow-up

The customer requests two options, then refers to the first one. The agent queries the CSVs and uses the price calculated in Python.

Customer

Quero duas opções de violões disponíveis até R$ 1.000. Mostre nome e preço.

Lia

Olá! Temos estas opções de violões disponíveis dentro do seu orçamento:

- Tagima Memphis AC-39 Nylon Natural - R$ 429,90 sem frete [D83].

- Rozini RC-104 Nylon Estudante - R$ 459,90 sem frete [D85].

Customer

E o primeiro no PIX?

Lia

O violão Tagima Memphis AC-39 Nylon Natural sai por R$ 408,41 sem frete no pagamento via PIX [D83].

Full export: catalog and PIX.

2. Promotion and non-stacking discounts

The answer combines a product lookup with the promotion policy. Presentation caveat: it does not explicitly state that the quoted price excludes shipping.

Customer

Quanto fica o Ohana CK-20 no PIX? Posso somar os 5% à promoção?

Lia

O Ohana CK-20 Concert Natural sai por R$ 439,20 no PIX (preço original de R$ 549,00 com 20% de desconto promocional) [D121].

Não é possível somar o desconto de PIX à promoção, pois as regras da loja determinam que as promoções não são cumulativas e o desconto de PIX não se aplica sobre preços já promocionais [P6.2-p6-1].

Full export: promotion and PIX.

3. Applying the installment policy

The agent retrieves the specific policy section and checks the minimum installment amount.

Customer

Uma compra de R$ 480 pode ser dividida em seis vezes sem juros?

Lia

Sim, uma compra de R$ 480,00 pode ser dividida em seis vezes sem juros, pois o valor da parcela (R$ 80,00) atinge o mínimo exigido para essa quantidade de parcelas [P3.1-p3-1].

Full export: installments.

4. Order status and tracking

Queried with **test customer #2** selected. The answer preserves the historical total and stored tracking code. The English word "shipped" inside the original Portuguese answer is a language caveat.

Customer

Qual o status, total e rastreamento do meu pedido 8?

Lia

O seu pedido 8 está com o status de enviado (shipped) e o código de rastreamento é BRJL5544332BR [O8]. O valor total histórico é de R$ 349,90 [O8].

Full export: order and tracking.

5. Out-of-scope request

The fixed service scope restricts the store to musical instruments. This conversation made no new tool call.

Customer

Vocês vendem cabos, palhetas ou amplificadores?

Lia

Olá! Na Empório da Música, nós somos especializados em instrumentos musicais, como violões, guitarras, baixos, teclados, baterias e outros instrumentos de cordas, sopro e percussão.

Cabos, palhetas e amplificadores são acessórios e não fazem parte do nosso catálogo de instrumentos.

Posso te ajudar a encontrar algum instrumento específico?

Full export: out of scope.

Review criteria and fixes

I guided tests around expected service behavior: source-supported facts, correct prices, context continuity, scope boundaries and restricted order access. I also defined expectations for tone and collecting the necessary information before advising on returns.

Automation and Codex supported execution. Reviewing answers and prioritizing fixes were part of my direction of the project. Three examples illustrate the process:

Observed issue	Implemented change	Recorded verification
Queries without customer identification or for another customer's order ended in citation errors.	Deterministic guidance based on the tool result, without revealing the order.	C24 and C27 passed targeted retests.
A discontinued product was described as out of stock despite a positive stored quantity.	Instructions distinguish commercial status from stock quantity.	C13 reported discontinued status and 13 units without offering a purchase.
An answer added "shipping excluded" to a historical order total.	Conservative validation and a regression test for the unsupported qualification.	C25 was repeated with correct status, total and tracking, without asserting a shipping breakdown.

These retests demonstrate the individual outcomes. Language and triage caveats remain, and the complete real-scenario suite was not rerun after the last fix.

Improvements with more time

I would start with response controls and interaction context. Query improvements would follow dataset growth, with quality evaluation after each change.

1. **Guardrails and triage:** validate price, stock and policy claims against tool results; structure missing triage fields and escalate uncertainty to staff. Add unseen questions and attempts to override the rules.
2. **Previous-interaction context:** retain summaries and relevant fields per identified customer, such as instrument preferences, budget and information already supplied. Access would follow authorization and retention rules; current prices, stock and policies would still be queried from sources.
3. **RAG for more documentation:** evaluate hybrid BM25/embedding retrieval followed by reranking. Improve context-preserving chunks, section metadata and version/effective-date controls. Compare retrieval and faithfulness on a separate question set.
4. **Queries over larger datasets:** replace CSV loading with databases or operational APIs, using filters, indexes and pagination. Keep tool parameters and permissions explicit, with pricing and access rules in code.
5. **Behavioral fine-tuning:** after improving data, prompts and validation, evaluate a compatible model using reviewed conversations for tone, triage and tool selection. Separate training from evaluation and measure gains against the base model. Prices and availability would remain source-backed queries.

Real authentication and stock/order integrations would be needed for operational use. These are proposed improvements; the submitted application remains a prototype.

Reports and examples: Gemini evaluation and five-conversation index. Implementation decisions are summarized in the README.